MONASH UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY

FIT5032 Internet Applications Development

Assessed Lab 3

Styling Web Application with CSS and Bootstrap in Vue.js 3

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Tutorial/Lab	Tuesday 2:00 pm
Project Name	`wzhang-library`

Tutor Name: Yujie Wang

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1 Lab Overview

This report presents the evidence completed for FIT5032 Assessed Lab 3. The lab focuses on styling a Vue.js 3 web application using CSS and Bootstrap 5. The existing library web application from the previous lab was extended with a user information form, Bootstrap form controls, submitted user information cards, and responsive layout support using Bootstrap breakpoint classes.

All assessed implementation work was completed individually. Peer interaction was limited to technical discussion and mutual support during the laboratory session.

2 Project Setup and Bootstrap Installation

2.1 Vue.js Project

The existing Vue.js 3 project named `wzhang-library` was reused and extended for the Week 3 laboratory activity. The development server was started using:

```
npm run dev
```

The application was tested in a browser to confirm that the Vue development server ran successfully.

2.2 Activity 1: Creating the Form Component

A new Vue component named `Form.vue` was created in the following directory:

```
src/components
```

The component was used to display the initial user information form title. The previous Lab 2 component was kept in the project but was commented out in `App.vue`, so that the new form component could be displayed for Lab 3.

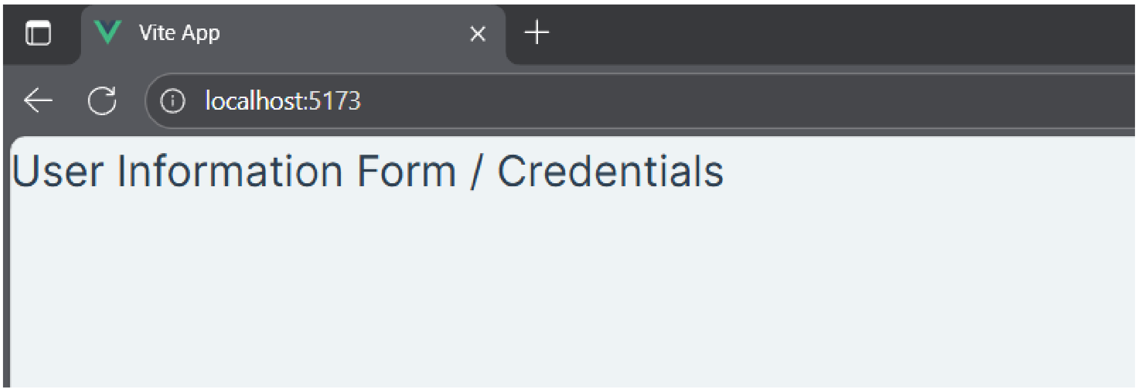

Figure 1: Activity 1: The newly created Form.vue component displayed in the running Vue application.

Brief description: Figure 1 shows that the new form component was successfully created and rendered in the Vue application.

2.3 Activity 4: Installing and Importing Bootstrap 5

Bootstrap 5 was installed in the project using the following command:

```
npm install bootstrap
```

The Bootstrap CSS file was then imported globally in `main.js`:

```
import 'bootstrap/dist/css/bootstrap.min.css'
```

Figure 2: Activity 4: Bootstrap 5 installed and imported in the Vue project.

Brief description: Figure 2 shows the Bootstrap import statement and confirms that Bootstrap styling was added to the project.

3 eFolio Task 3.1: CSS and Bootstrap Form Implementation

3.1 Activity 2: Adding the Basic HTML Form

A user information form was added to `Form.vue`. The form included the required input elements: username, password, Australian resident status, reason for joining, and gender selection.

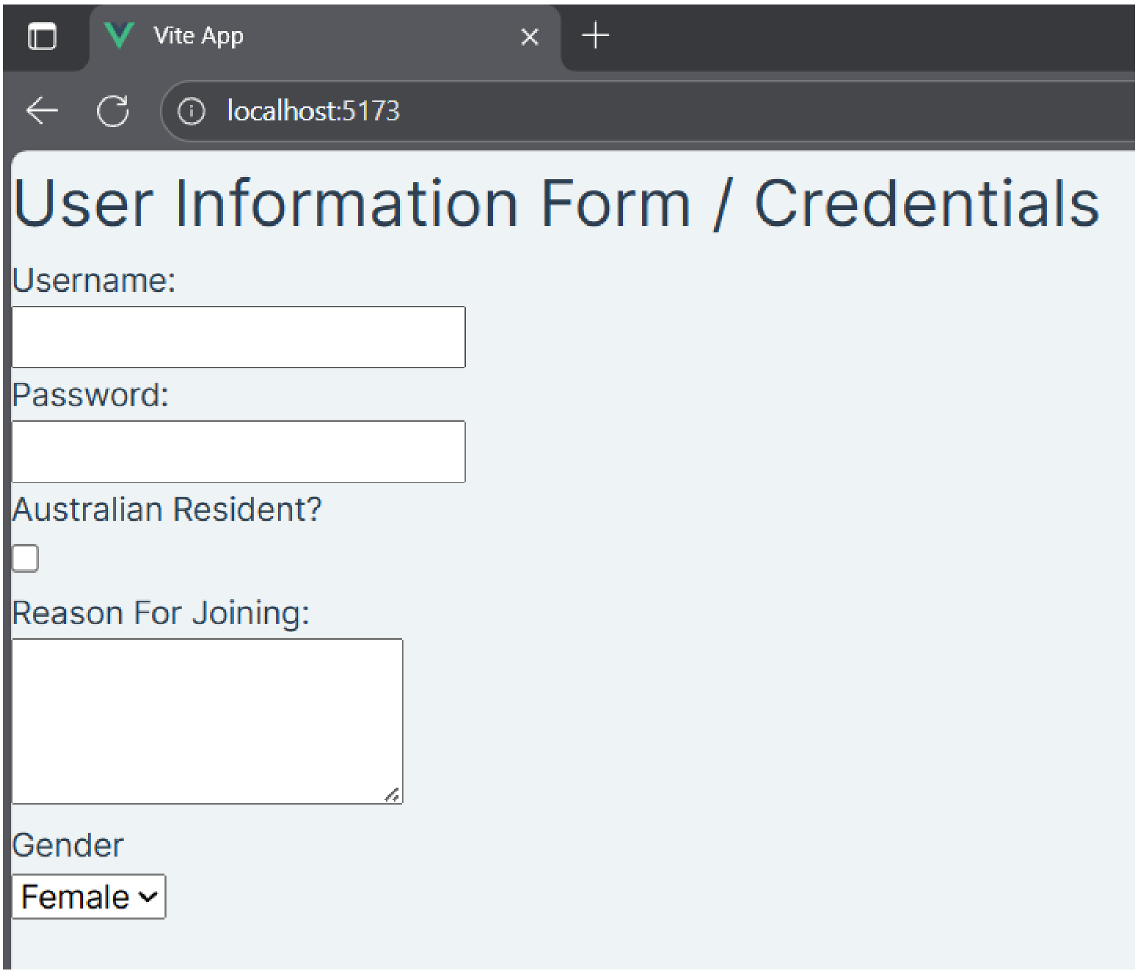

Figure 3: Activity 2: Basic HTML user information form before Bootstrap styling.

Brief description: Figure 3 shows the basic form structure before Bootstrap styling was applied.

3.2 Activity 3: Styling the Form with CSS

A separate CSS file was created and used to practise different CSS selector types. The form title, form container, focused input elements, and gender selection field were styled using tag, class, ID, and attribute selectors.

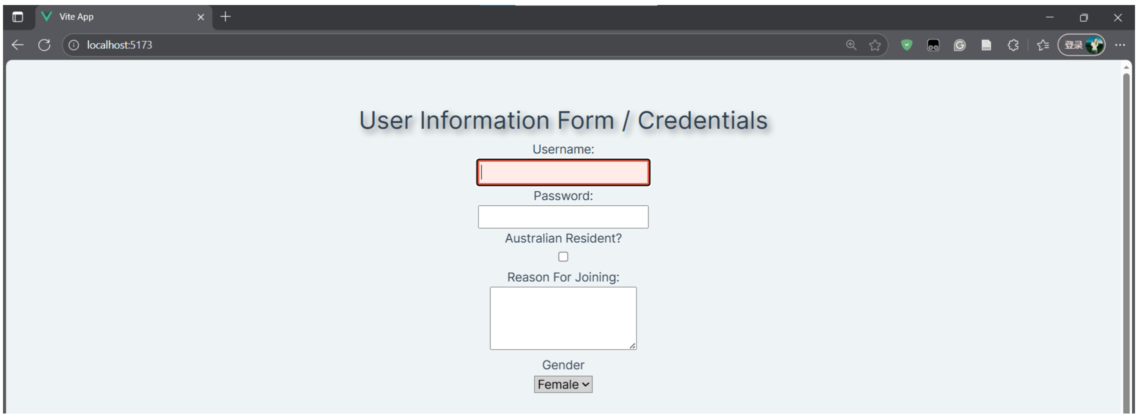

Figure 4: Activity 3: User information form styled with regular CSS selectors.

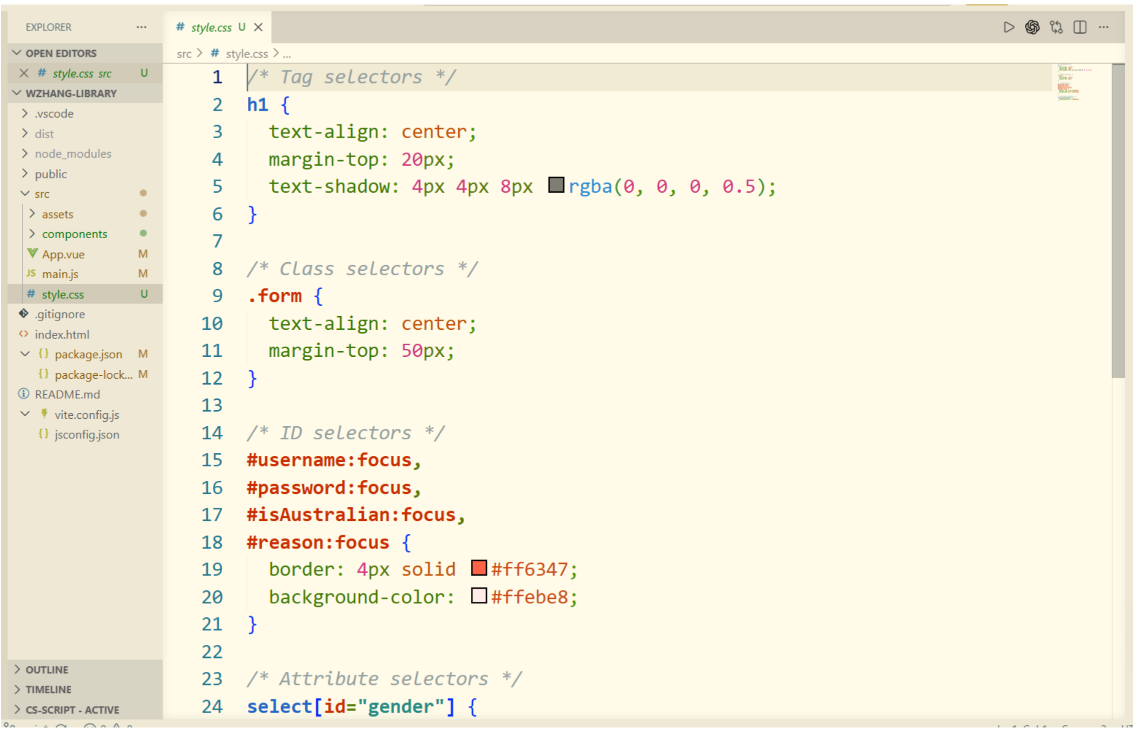

Figure 5: Activity 3: CSS code showing tag, class, ID, and attribute selectors.

Brief description: Figures 4 and 5 show the regular CSS styling result and the corresponding CSS selector implementation.

3.3 Activity 5: Styling the Form with Bootstrap 5

The form was then refactored using Bootstrap 5 classes. Bootstrap layout and form classes were used to improve spacing, alignment, readability, and visual consistency.

The main Bootstrap classes used included:

- `container`, `row`, and column classes for layout;
- `mb-3` for spacing between form fields;
- `form-control` for text inputs and textareas;
- `form-check`, `form-check-input`, and `form-check-label` for the checkbox;
- `form-select` for the gender selection field;
- `btn`, `btn-primary`, and `btn-secondary` for the form buttons.

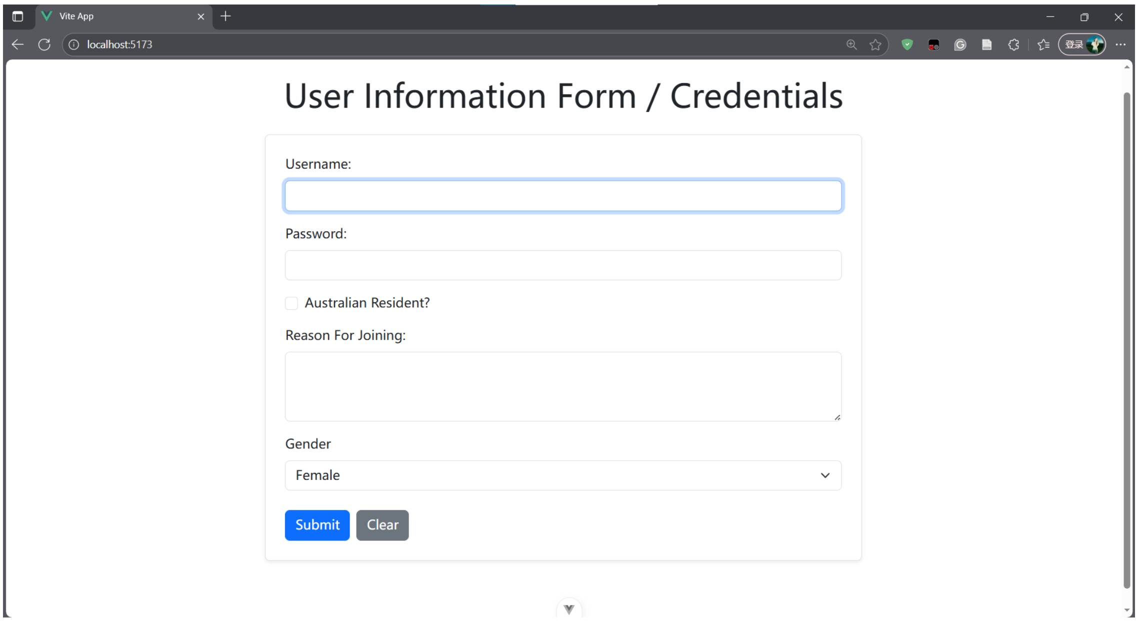

Figure 6: Activity 5: User information form styled with Bootstrap 5 classes.

Figure 7: Activity 5: Source code showing Bootstrap 5 classes used in the form.

Brief description: Figures 6 and 7 show the completed Bootstrap-powered form and the source code used to implement it.

3.4 Activity 6: Displaying Submitted User Information

Vue's `ref` function and `v-model` bindings were used to store form data reactively. The form submission was handled using `@submit.prevent`, which prevents the default browser submit behaviour and calls the custom submit function.

Submitted user information was displayed below the form using Bootstrap cards. Each submitted record was rendered through `v-for` and displayed as a separate card.

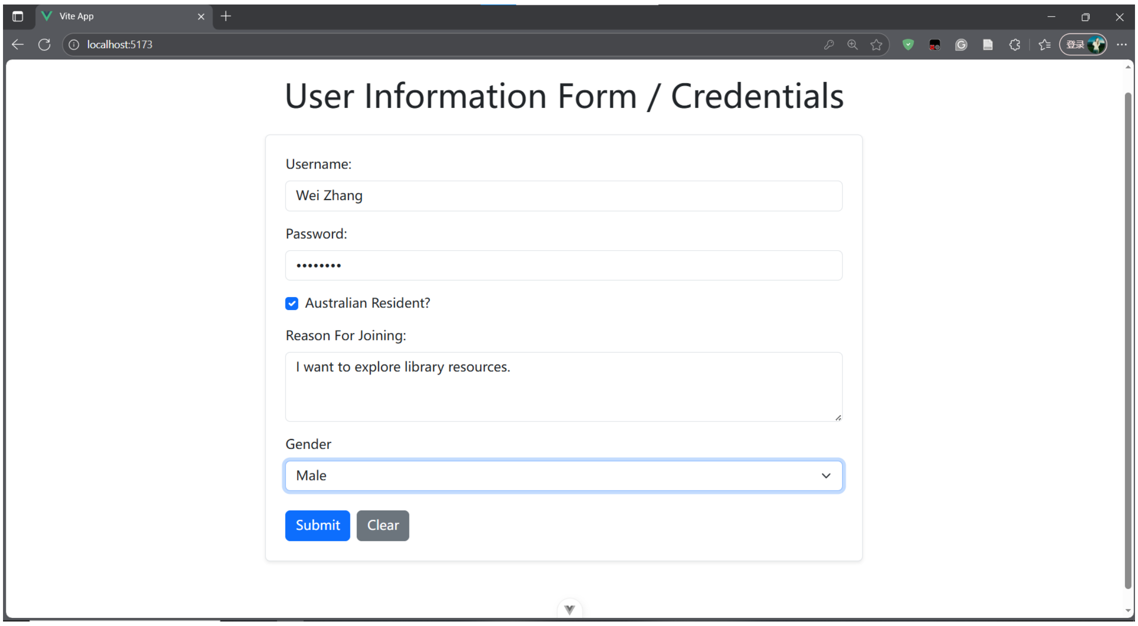

Figure 8: Activity 6: The Bootstrap form filled with user information before submission.

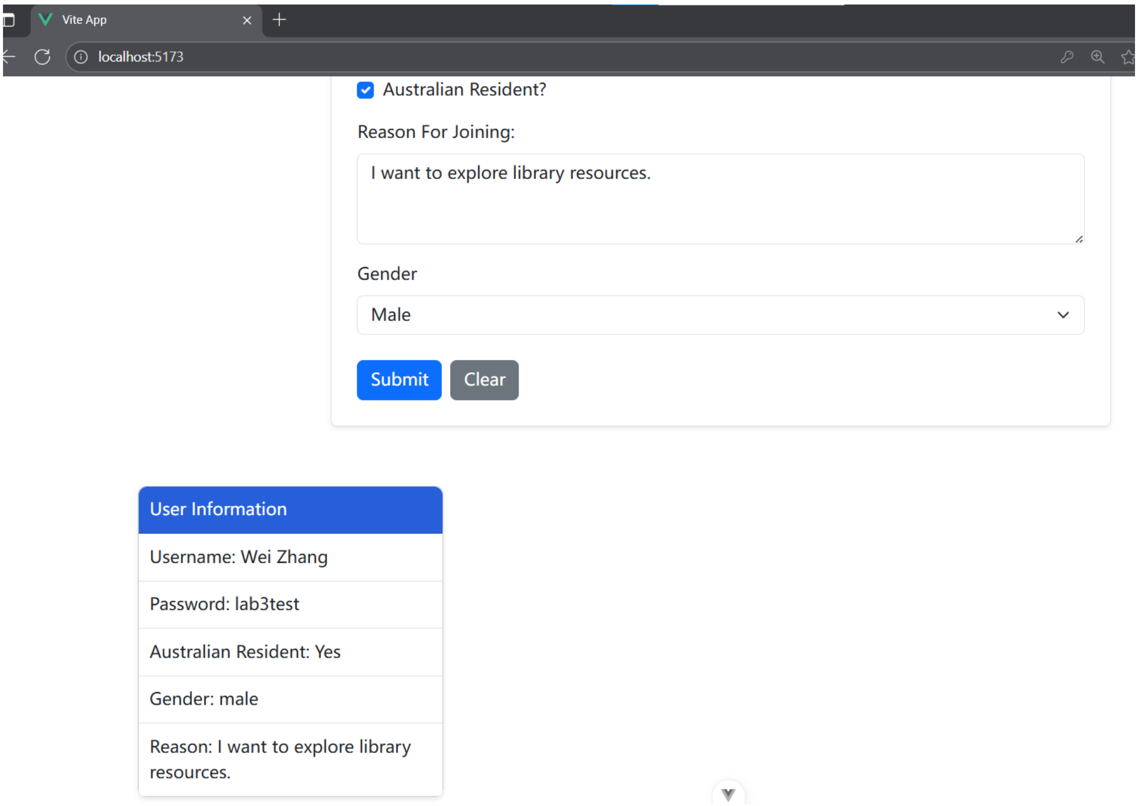

Figure 9: Activity 6: Submitted user information displayed using Bootstrap cards.

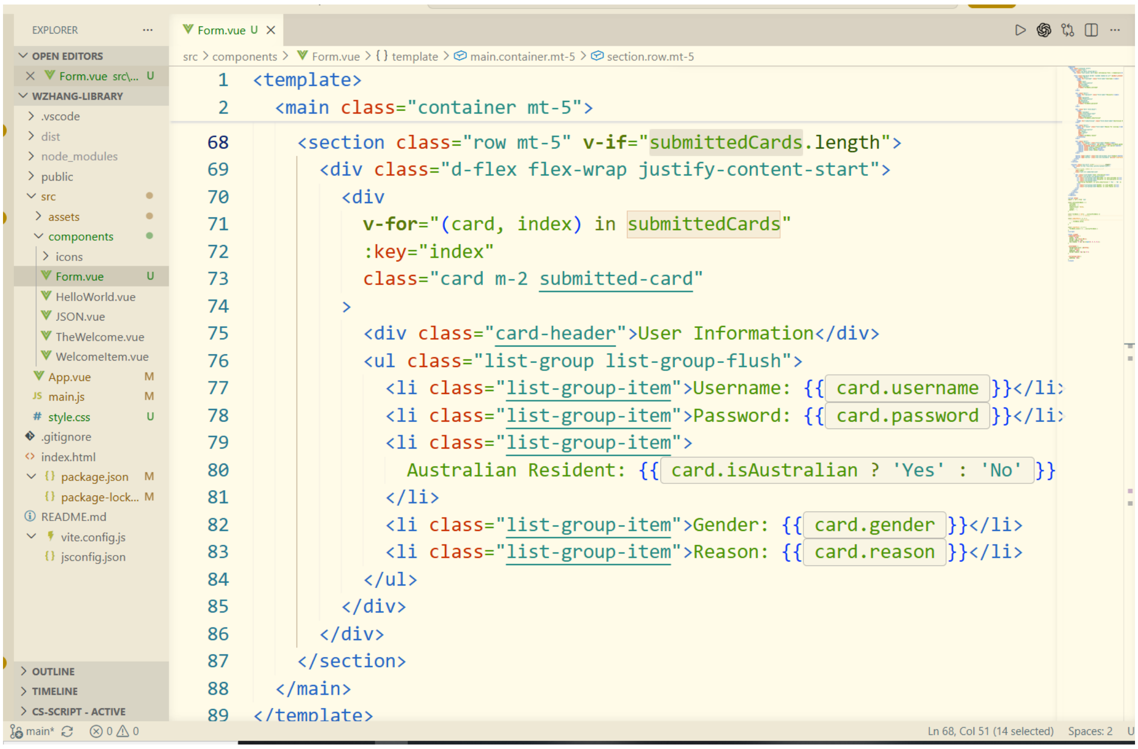

Figure 10: Activity 6: Source code implementing formData, submittedCards, and submitForm.

Brief description: Figures 8, 9, and 10 show the completed submit functionality and the Bootstrap card output.

3.5 Clear Button Functionality

The clear button was implemented using a custom `clearForm` function. When the button is clicked, all form fields are reset to their initial values.

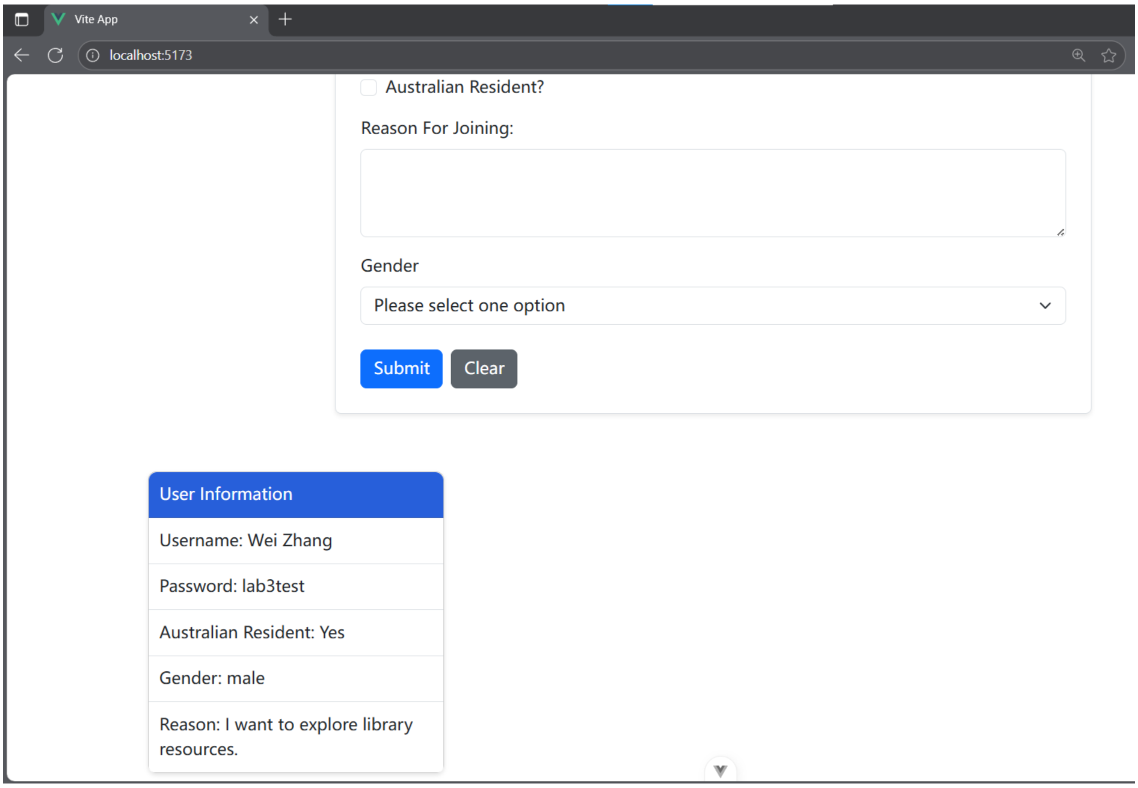

Figure 11: Activity 6: The form after the Clear button resets all input fields.

Figure 12: Activity 6: Source code implementing the Clear button functionality.

Brief description: Figures 11 and 12 show that the Clear button resets the form fields successfully.

4 eFolio Task 3.2: Responsive Design and GitHub Evidence

4.1 Activity 7: Responsive Form with Bootstrap Breakpoints

Bootstrap breakpoint classes were introduced to make the user information form support different devices. The form layout was adjusted using responsive column classes so that the form remains readable and usable on smartphones, tablets, laptops, and larger screens.

The responsive implementation used Bootstrap breakpoint classes such as:

```
col-12 col-sm-10 col-md-8 col-lg-6 col-xl-5 mx-auto
```

This allows the form to occupy a wider area on smaller screens while remaining centred and visually balanced on medium and large screens.

Figure 13: Activity 7: Source code showing Bootstrap breakpoint classes used for responsive layout.

Brief description: Figure 13 shows the Bootstrap breakpoint classes used to support different device sizes.

4.2 Responsive Design Evidence

The responsive behaviour was tested using the browser device toolbar. Screenshots were taken at different viewport sizes to demonstrate that the form remains usable across devices.

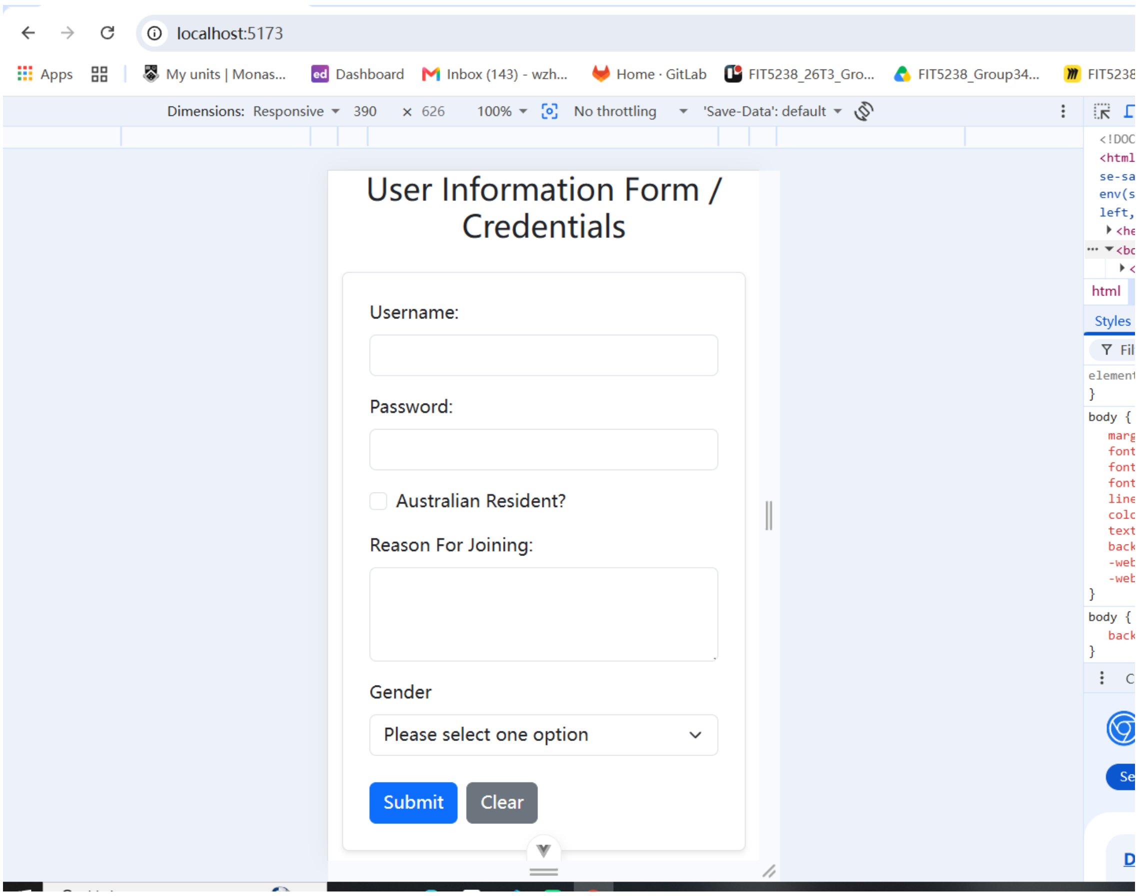

Figure 14: Activity 7: User information form displayed on a smartphone-sized viewport.

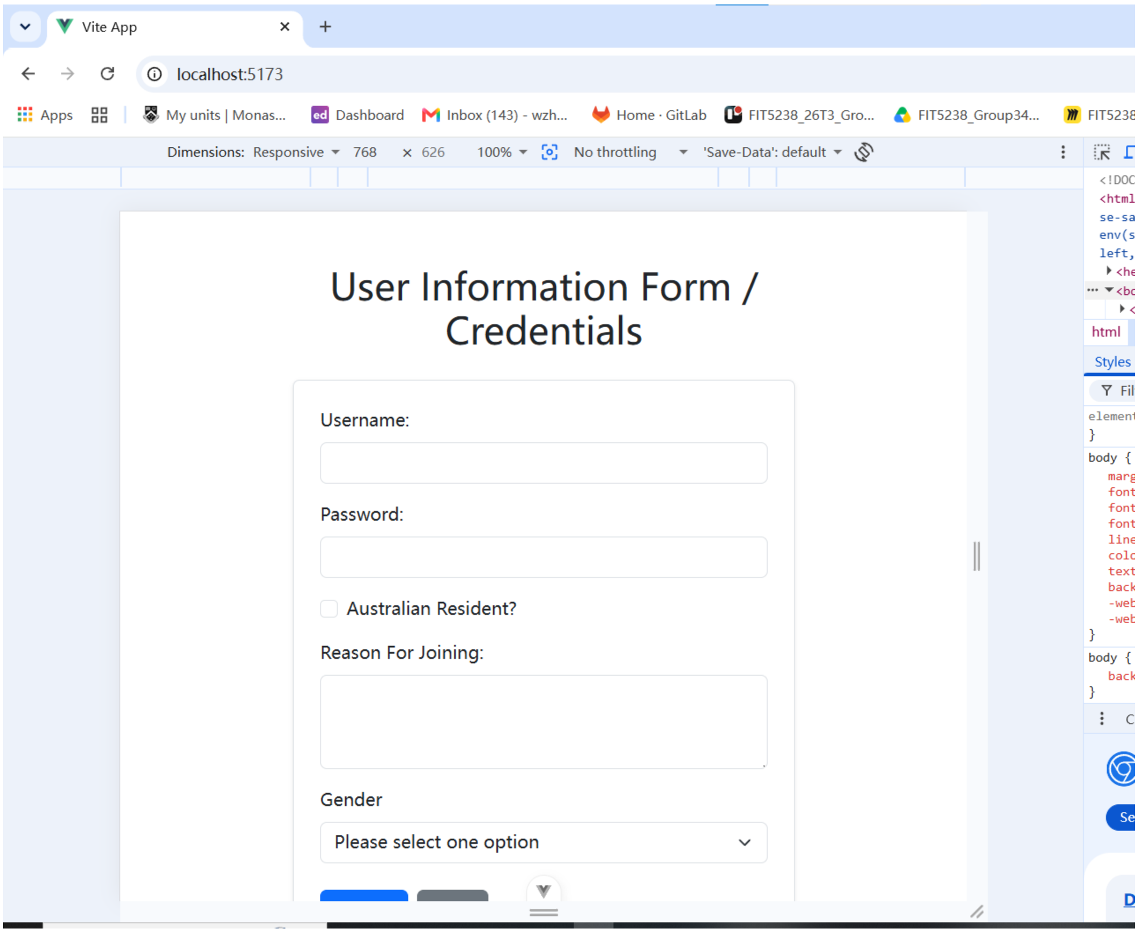

Figure 15: Activity 7: User information form displayed on a tablet-sized viewport.

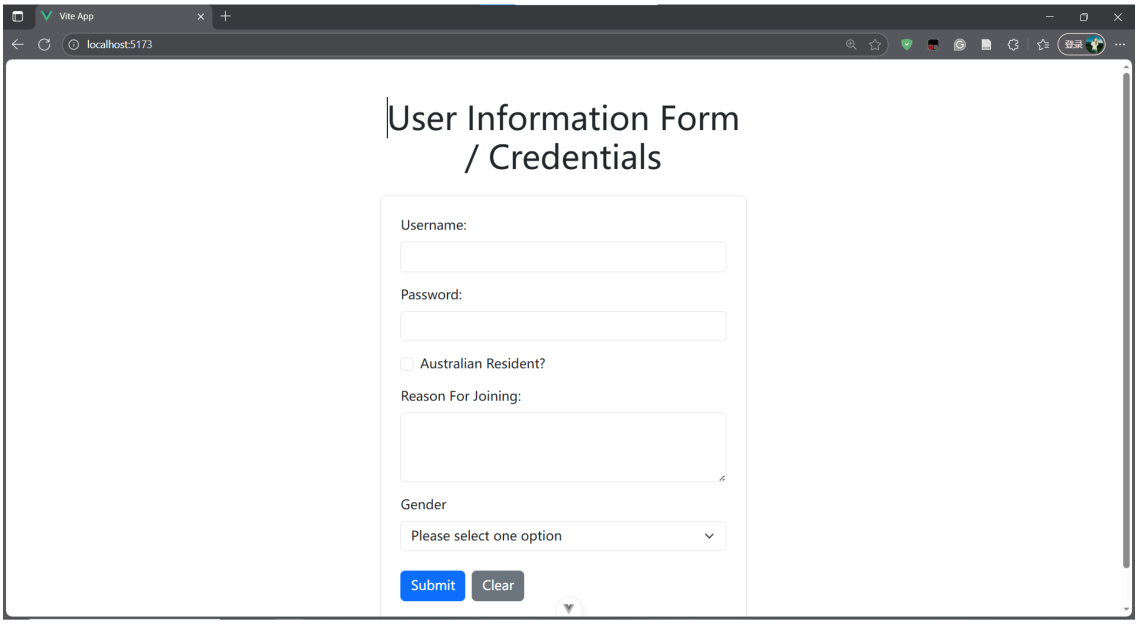

Figure 16: Activity 7: User information form displayed on a laptop or desktop viewport.

Brief description: Figures 14, 15, and 16 show that the Bootstrap-powered form supports different screen widths.

5 GitHub Version Control Evidence

The completed Vue.js project was committed and pushed to a private GitHub repository for version management and maintenance. The repository was shared with the tutor as required.

Project Name	`wzhang-library`
GitHub Repository	`https://github.com/weiwei0618seu/wzhang-library`

Figure 17 shows the GitHub commit history after the Lab 3 project update was pushed successfully.

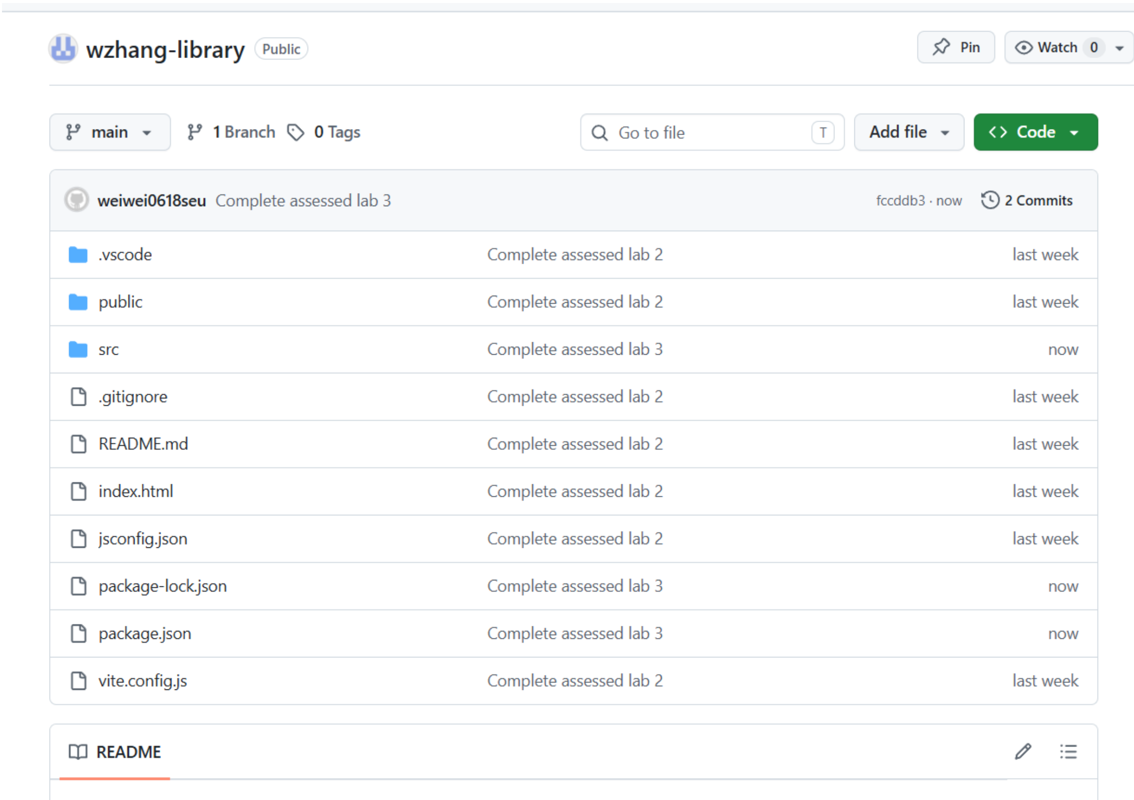

Figure 17: GitHub commit history showing the successful Lab 3 project update.

6 Evidence Checklist

Assessment Item	Evidence Included	Status
Activity 1	Screenshot showing the newly created `Form.vue` component rendered in the Vue application	Complete
Activity 2	Screenshot showing the basic HTML user information form	Complete
Activity 3	Screenshots showing the regular CSS styling result and CSS selector implementation	Complete
Activity 4	Screenshot showing Bootstrap 5 installed and imported in `main.js`	Complete
Activity 5	Screenshots showing the Bootstrap-powered form and the related Bootstrap form code	Complete
Activity 6	Screenshots showing filled form data, submitted Bootstrap cards, submit code, and Clear button functionality	Complete
Activity 7	Screenshots showing responsive form support on mobile, tablet, and laptop viewports, plus breakpoint code	Complete
GitHub History	Screenshot showing the successful GitHub push and commit history	Complete

Table 1: Assessed Lab 3 evidence checklist.

7 Completion Summary

The required tasks for FIT5032 Assessed Lab 3 were completed successfully. The existing Vue.js library application was extended with a user information form. The form was first created using basic HTML form elements, then styled using regular CSS selectors, and finally refactored with Bootstrap 5 classes. Vue data binding was used to collect form data, and submitted information was displayed using Bootstrap cards. The Clear button was implemented to reset the form fields. Bootstrap breakpoint classes were also used to make the form responsive across different devices. Finally, the completed project was committed and pushed to GitHub, and the required evidence has been included in this report.